ASSIGNMENT #2. WHEN ARE STAR BASKETBALL PLAYERS MOST EFFECTIVE?

Background

The use of data and analytics within professional sports has grown considerably in the past several years. This allows for more data driven and informed decision making and allows sport professionals to challenge assumptions that have often been held by coaches and players. In that spirit, I would like to use play-by-play (NBA) basketball data to provide some analytics on what makes certain players or teams successful (I will highlight the specific questions and data points that I am asking you to answer below).

Some of you will find this exciting just because you have an interest in basketball. However, if you do not, you may ask yourself why we are using this data and context for this assignment. Sports provides a context where the client (your managers, the audience in your presentation, etc.) will know more about the empirical context than you do. Therefore, you have to be careful and data driven making sure that your assumptions and arguments are consistent with what your audience expects. If you do not feel comfortable with this, try to first get some background on basketball (either watching games, watching analysts discussing games, etc.). You can also feel free to reach out to me, and make sure that you understand things correctly.

As before, this assignment is unstructured and it is up to you to define how you will do the analysis, and how you will define various metrics. Start from the goal you are trying to reach, and work towards how you can provide an answer with the data you have been given.

Assignment

Part A) *How does the play of a star basketball player change depending on the teammates they are playing with?*

The composition of a basketball team is inherently fluid. Across seasons, teams add new players and try to form partnerships between teammates.

For example, the Golden State Warriors added Kevin Durant, who played alongside Steffan Curry. The Milwaukee Bucks added Damian Lillard alongside Giannis Antetokounmpo. Kyrie Irving came to the Dallas Mavericks to play alongside Luka Doncic, and was also joined by Klay Thompson. The addition of another "star" player to a team, or the movement of a "star" player between teams is a big shift in the composition of a basketball team.

Focus your analysis on a player that is moving to a new team and how their play changes, or on the team they are joining and how the existing star players adjusted their play to this new player.
You may focus on a single player or "star" players in general. You may focus on a single player and demonstrate how their play changed with different team-mates, or you may choose in broad terms how several leading players change in response to the composition of their teams.

It is up to you to consider the players that you want. It is also up to you to consider how you would define success in this context. You need to decide which player performance metrics you are considering, and which data frame you will look at. For example, are players taking different types of shots? Are they more efficient when shooting? Are they able to pass the ball more easily? Are the teams more able to rest star players because they have another star player they can play instead.

The point of the assignment is to be clear in articulating your logic and providing appropriate evidence from the data. You do not need to use complex analytics (e.g. elaborate ML to predict an outcome). You can complete this assignment with the pivot tables we learned in Workshop #3. But you have to think through your data, metric selection, and empirical comparisons.

If you are not comfortable with basketball, please use AI tools to gain an understanding and develop a direction. *Ex. https://claude.ai/share/7a1c97e1-459e-49dc-a698-07c530f93c24*

Part B) *What is the impact of adding a "star" player to the performance of a team?*

The final result of a basketball game is a function of many factors. Yet, the addition of a star player may lead to a subtle increase in performance that can potentially lead to more victories. However, it is often not clear how much adding a new player increases the performance of a team, and specifically quantifying the increase in performance.

You can select either a focal team, or several teams that traded for star players. Then you can develop a way of quantifying the aggregate performance of the team and how / when the acquisition of the star player helped, and how/when it hurt the team. You have considerable flexibility regarding how you might define and measure what constitutes aggregate team performance. You can look at performance metrics such as +/- (plus minus) and compute how this metric improved? You can calculate how many points are scored by different players and whether this addition increases the performance of some players, while limiting the performance of others.

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Deliverable

Summary slides with approximately 2 to 3 slides summarizing your approach to each. Do not show python code but attempt to show visualizations or clear figures summarizing your findings. You can include an appendix file with code or any other information you find helpful. If you would like, I would be happy to provide you with feedback on your approach early on enough in the assignment.

Data

I will be providing you with play-by-play data from several NBA seasons. You can only focus on a narrow sample of data and firms. The key is to select the data that can be the most useful for your specific focus.

Due Dates

- Part A&B) Bonus - *Friday March 28, 2025 @ 5pm; Regular Submission - Friday April 4, 2025 @ 5pm*

Grading

Grade Component	Weight of Total Grade
Part A) *How does the play of a star basketball player change depending on the teammates they are playing with?*	12.5%
Part B) *What is the impact of adding a "star" player to the performance of a team?*	12.5%

Grade Rubrics	Share of Assignment
Have you selected a reasonable question	10%
Empirical Approach (*Do you have a plausible approach to answer the question?*)	40%
Insights (*Do you arrive at meaningful and informative insights?)*	40%
Style (*Is it clear and well explained?*)	*10%*